

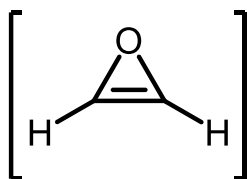
3. Kleine Ringe

3.1. Dreiring-Heterocyclen

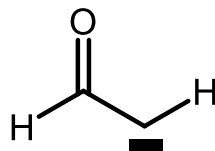
Ungesättigte (Oxiren, Thiiren, 2*H*-Azirin, 3*H*-Diazirin) und gesättigte Systeme (Oxiran, Dioxiran, Aziridin)

3.1.1. Ungesättigte Dreiringheterocyclen

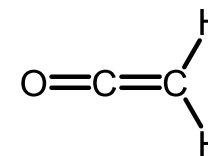
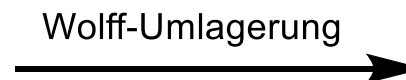
3.1.1.1. Oxiren



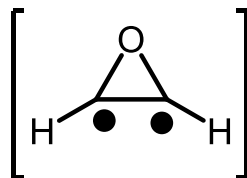
4π-Antiaromat
Singulett-
Oxiren
(unbekannt)
45 kcal/mol



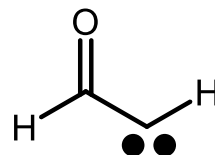
Singulett-Formylcarben
44 kcal/mol



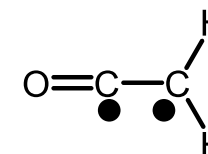
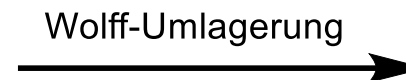
Singulett-Keten
-34 kcal/mol



Triplett-Oxiren
(unbekannt)
69 kcal/mol



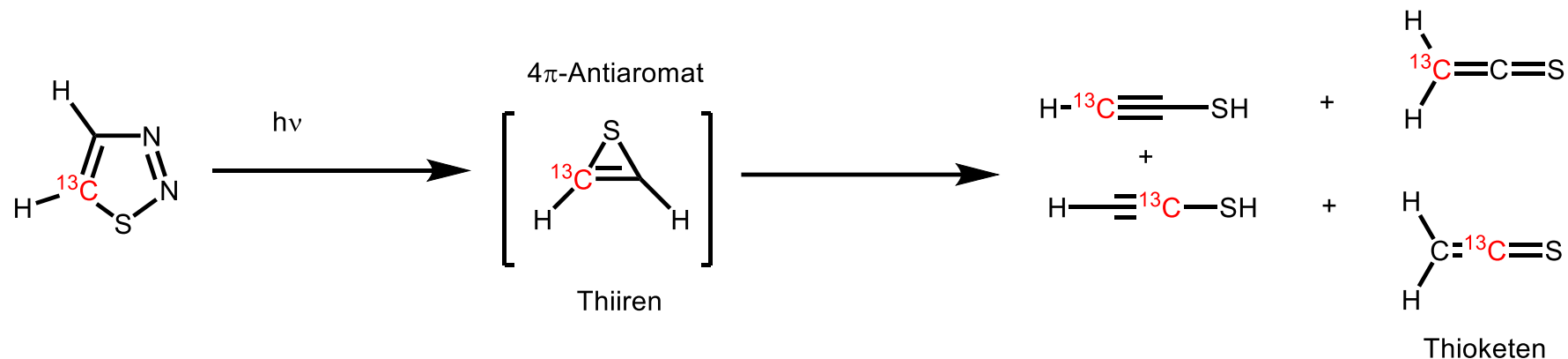
Triplett-Formylcarben
41 kcal/mol



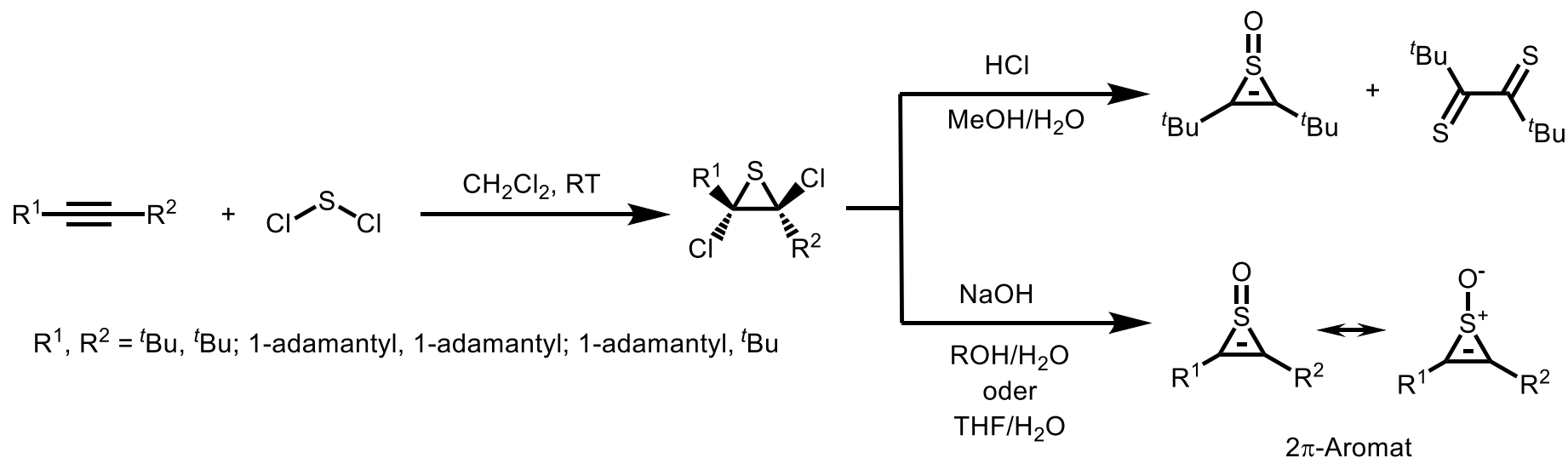
Triplett-Keten
20 kcal/mol

3.1.1.2. Thiiren

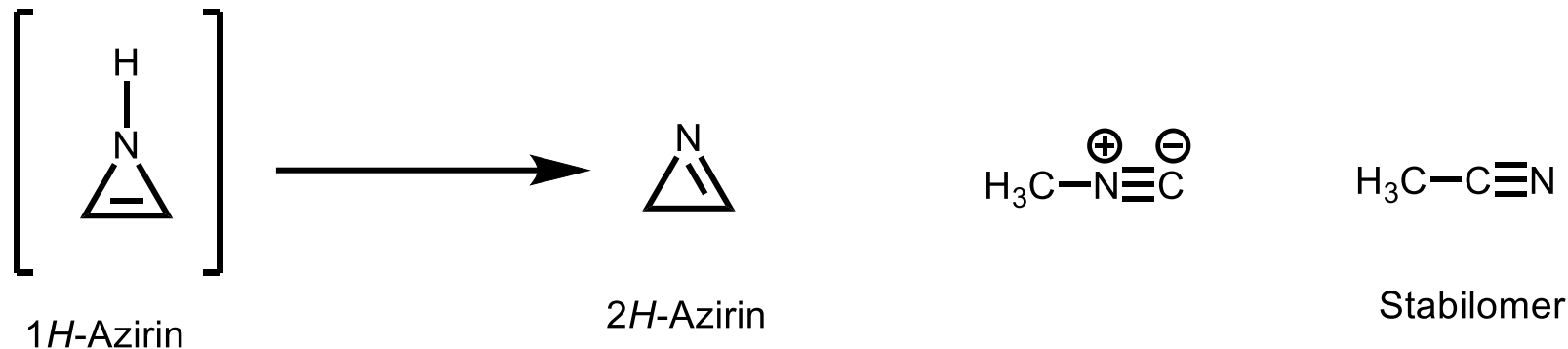
Matrixisolation, Isotopen: A. Krantz, J. Laurenzi, *J. Am. Chem. Soc.* **1981**, 103, 486.



Thiiren-S-oxid: J. Nakayama, K. Takahashi, Y. Sugihara, A. Ishii, *Tetrahedron Lett.* **2001**, 42, 4017.



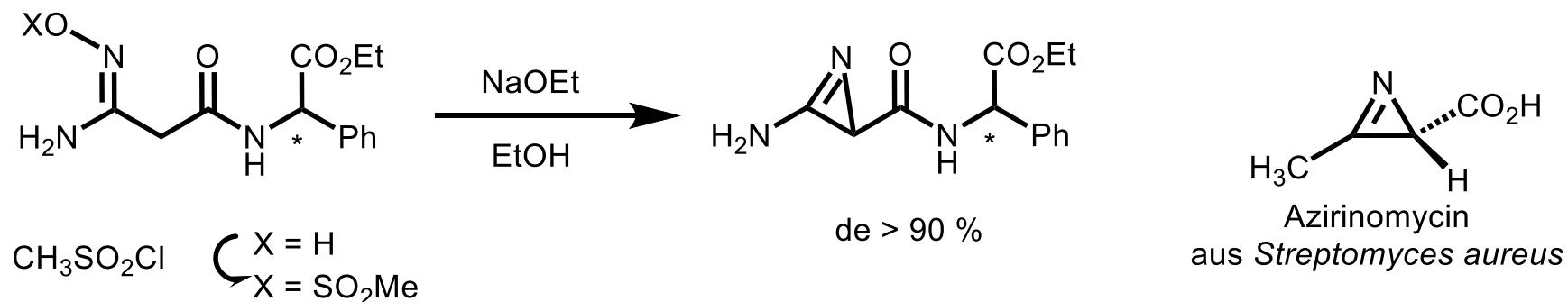
3.1.1.3. 2*H*-Azirin



Das antiaromatische 4π -System kann ohne Ringöffnung umgangen werden:
 Abwinklung von 1*H* oder besser durch Tautomerisierung zum 2*H*-Isomer
 Spannungsenthalpie von 2*H*-Azirin $\sim 190 \text{ kJmol}^{-1}$ (vgl. Cyclopropen $\sim 230 \text{ kJmol}^{-1}$)

Synthesen

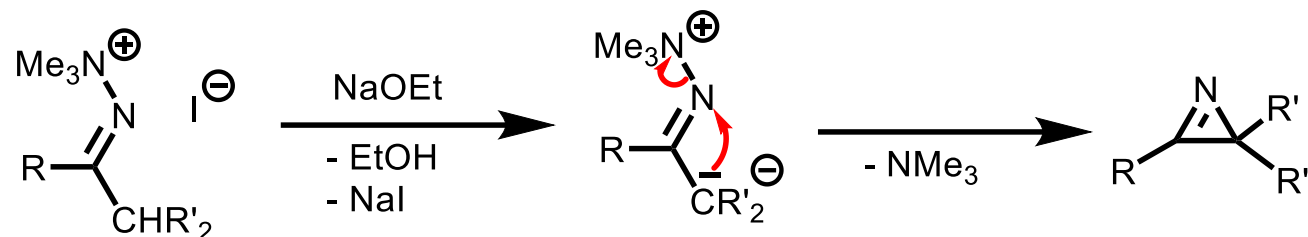
a) Modifizierte Neber-Reaktion



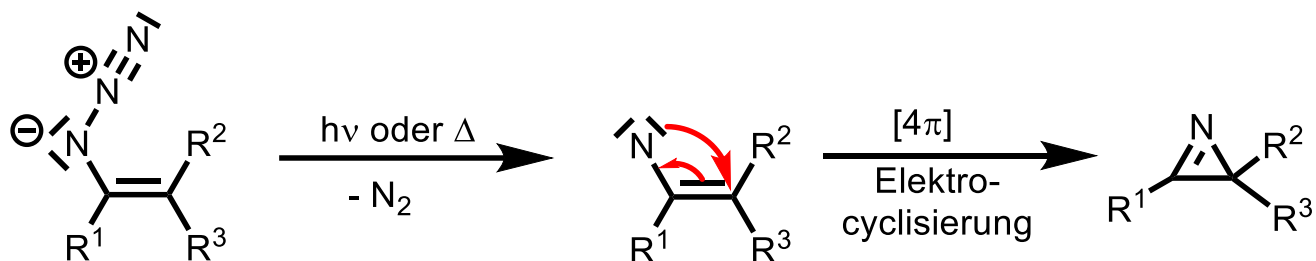
3.1.1.3. 2H-Azirin

Fortsetzung

b) Aus methylierten Dimethylhydrazonen

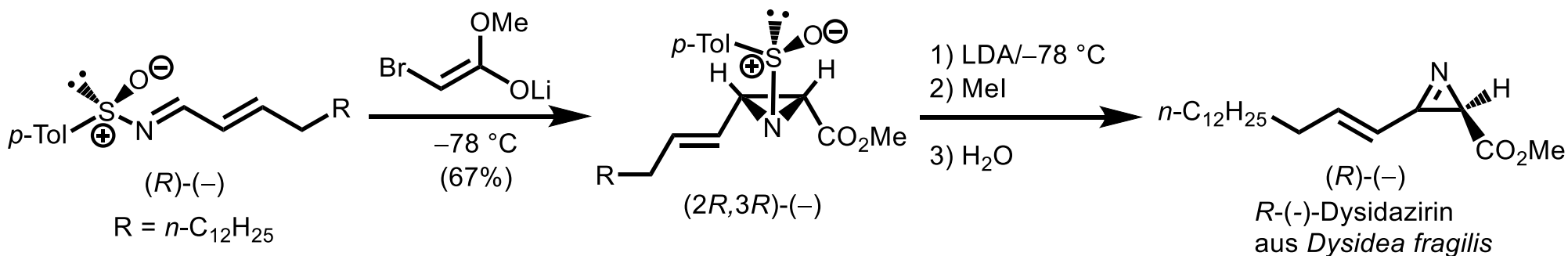


c) Thermo- oder Photolyse von Alkenylaziden



d) Eliminierung aus Aziridin

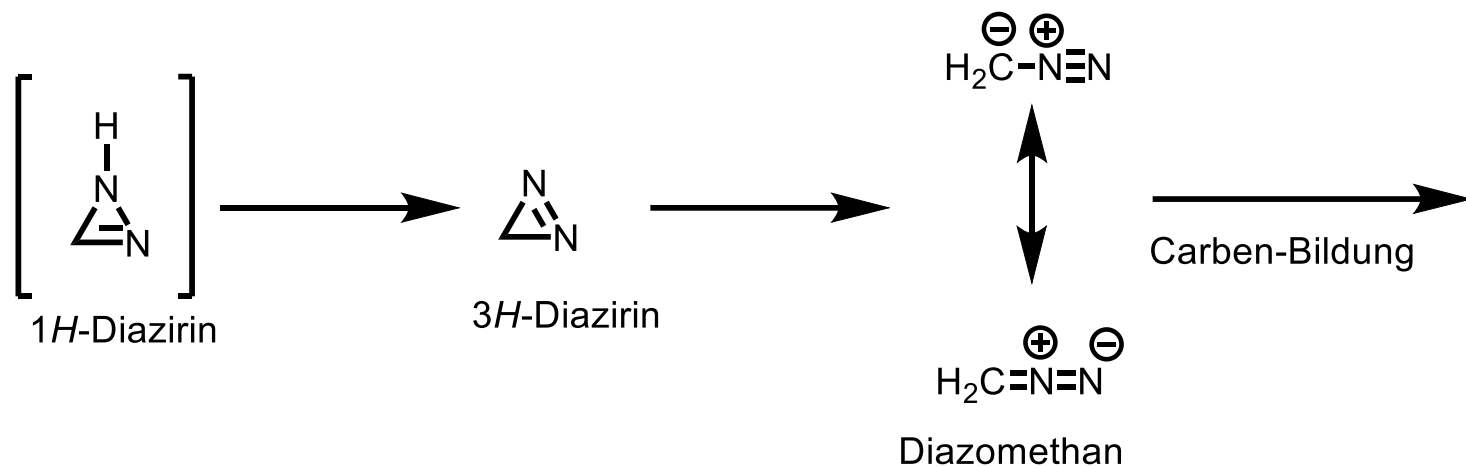
Asymmetrische Synthese, Cytotoxisches Antibiotikum (R)-(-)-Dysidazirin:
F. A. Davis, G. V. Reddy, H. Liu, *J. Am. Chem. Soc.* **1995**, 117, 3651.



(R)-(-)-Dysidazirin
aus *Dysidea fragilis*

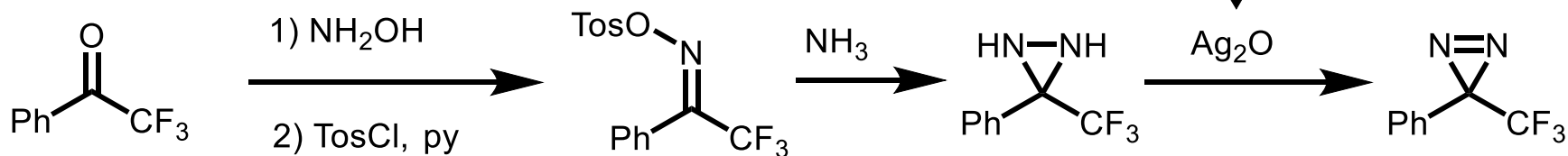
3.1.1.4. 3H-Diazirin

13



Synthese

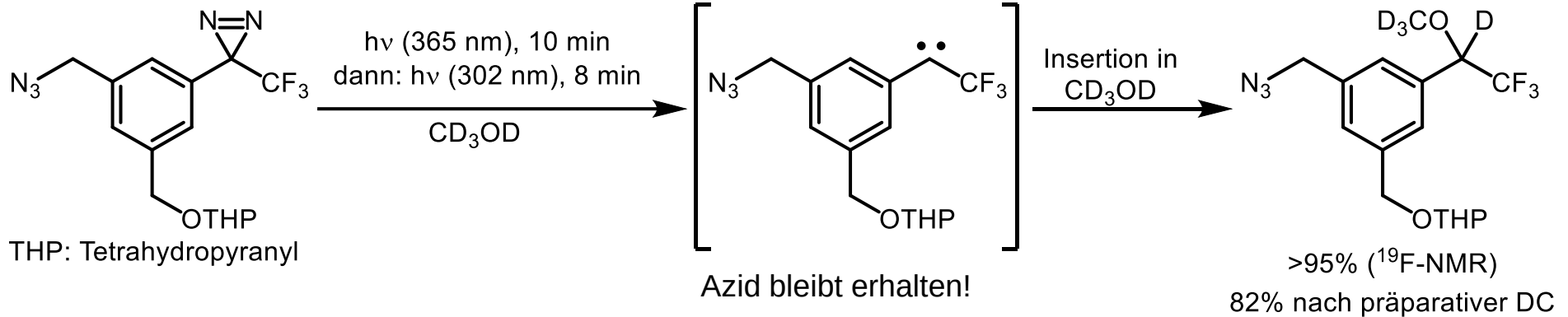
Brunner, Richards, *J. Biol. Chem.* **1980**, 255, 3313



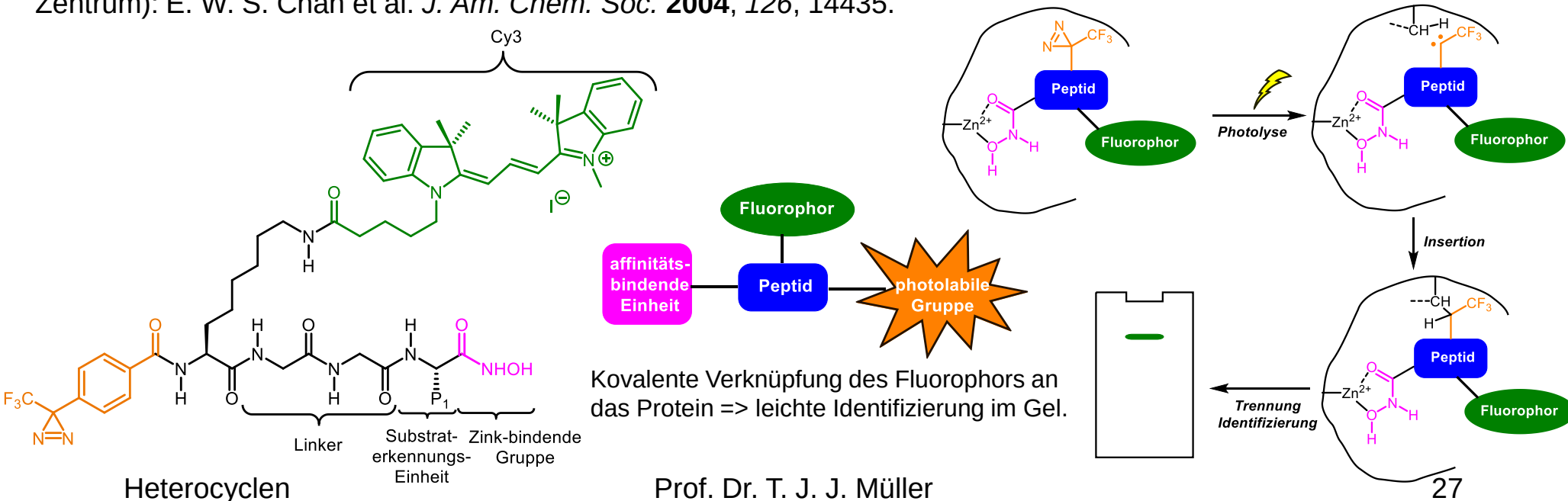
3.1.1.4. 3H-Diazirin

Fortsetzung

3H-Diazirine sind Carbenquellen 3H-Diazirine in der Photoaffinitätsmarkierung: T. Hosoya et al., *Org. Biomol. Chem.* **2004**, 2, 637.



Photoaffinitätsmarkierung von Proteinen (hier selektive Hydroxamat-Sonde für Metalloproteasen mit Zn²⁺ im aktiven Zentrum): E. W. S. Chan et al. *J. Am. Chem. Soc.* **2004**, 126, 14435.



3.1.2. Gesättigte Dreiringheterocyclen

Exkurs: Ringspannung

"Baeyer-Spannung(senthalpie)" (1885): Abweichung der Verbrennungsenthalpie pro Ringposition vom Fall des Cyclohexans.

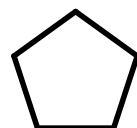
14



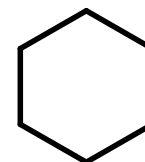
38.5



27.2



5.4



0

[kJ/mol]

Absolutwert der Verbrennungswärme pro CH₂-Gruppe bei Cyclohexan: 658.6 kJ/mol

Substituent X				
	0.2 0.0	6.4 6.2	25.7 26.5	27.3 27.5
	1.2 0.0	5.7 5.4	24.9 24.7	26.4 26.3
	1.1 0.0	6.1 5.8	25.4 25.2	27.0 26.7

Spannungsenthalpien
in kcal/mol (1 kcal/mol = 4.184 kJ/mol)
(siehe *JOC* **2002**, 3488)

berechnete und **experimentell**
bestimmte Werte

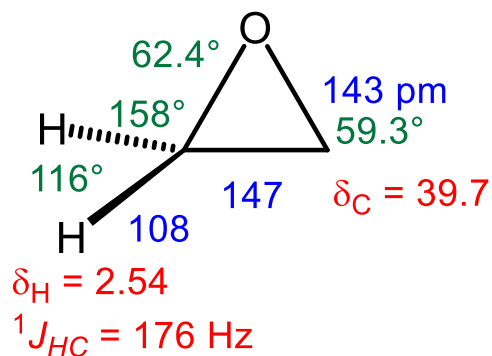
Ringspannung der 3- und 4-
Ringe sehr ähnlich:
Heteroatom haben kaum einen Einfluss

3.1.2. Gesättigte Dreiringheterocyclen

Fortsetzung

Steckbrief von Oxiran, Aziridin und Thiiran

Oxiran



Spannungsenthalpie 114 kJ/mol

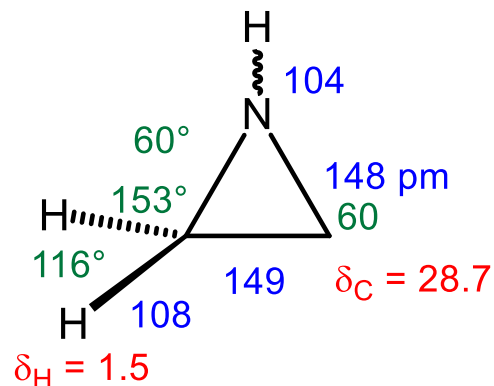
Sdp 11 °C; Produktion: $7 \cdot 10^6$ t/a

Ionisierungsenergie: 10.5 eV
 (Elektron aus nichtbindendem Elektronenpaar des O);

$\delta (^1\text{H}) = 2.54 \text{ ppm}$, $\delta (^{13}\text{C}) = 39.7 \text{ ppm}$;

Brönsted-Base, Lewis-Base

Aziridin



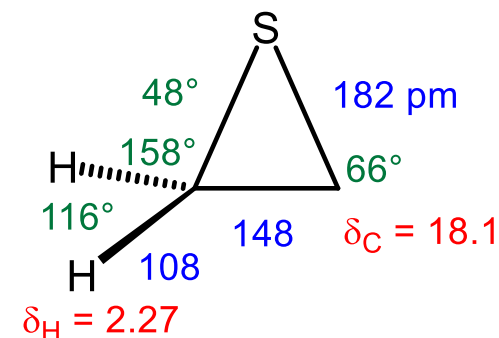
113 kJ/mol

Sdp 57 °C

Ionisierungsenergie: 9.8 eV

$\delta (^1\text{H}) = 1.5, 1.0 \text{ ppm}$, $\delta (^{13}\text{C}) = 28.7 \text{ ppm}$

Thiiran



83 kJ/mol

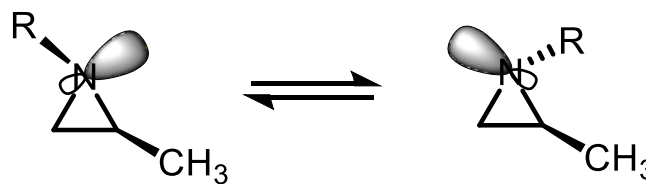
Sdp 55 °C

Ionisierungsenergie: 9.05 eV

$\delta (^1\text{H}) = 2.27 \text{ ppm}$, $\delta (^{13}\text{C}) = 18.1 \text{ ppm}$

farblose, in Wasser kaum lösliche Flüssigkeit

15

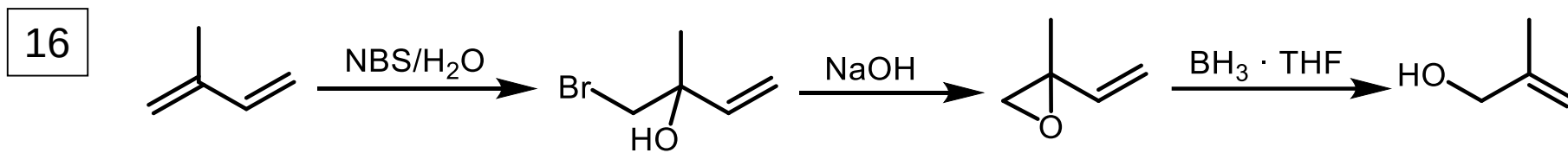


pyramidale Stickstoff-Inversion: R = Me: 70 kJ/mol; R = Cl: 112 kJ/mol

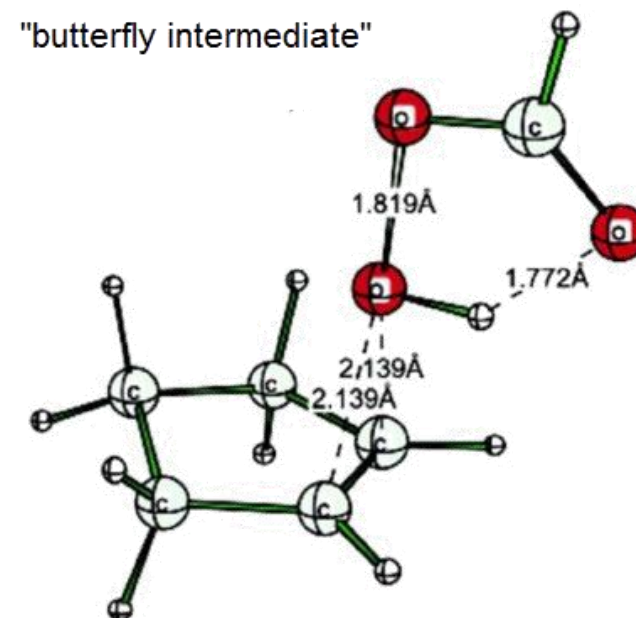
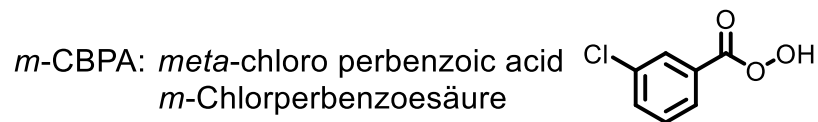
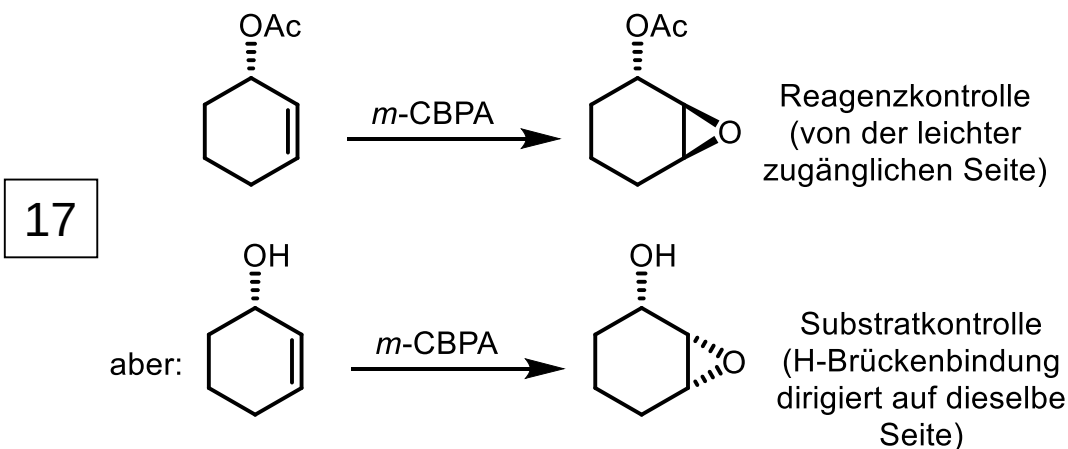
3.1.2.1. Oxiran

Synthesen

a) Aus Halohydrinen mit Basen (elektronenreiche Alkene reagieren rascher)



b) Aus Alkenen mit Persäuren (Prilezhaev-Reaktion)

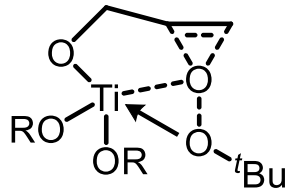


Berechnung des ÜZ: R. D. Bach, O. Dmitrenko, W. Adam, S. Schambony, *J. Am. Chem. Soc.* **2003**, 125, 924.

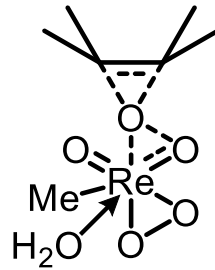
3.1.2.1. Oxiran

Fortsetzung

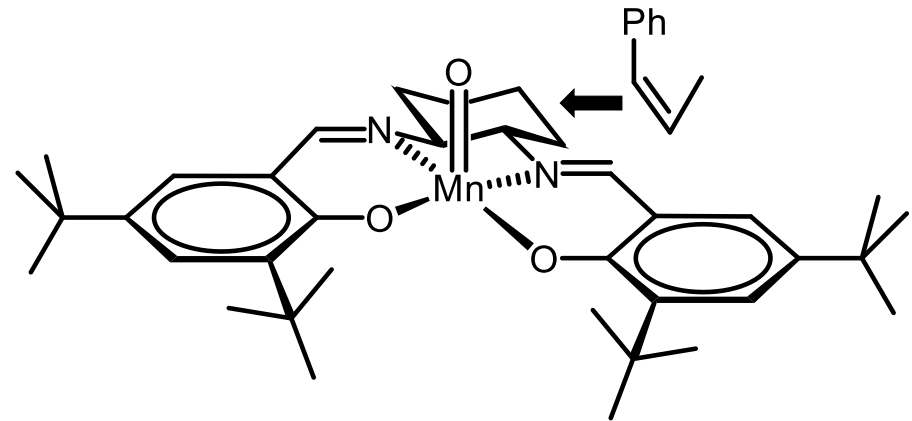
c) Übergangsmetallkatalysiert



Peroxy-Komplex
Sharpless-Epoxidierung



Peroxo-Komplex
Herrmann-Epoxidierung



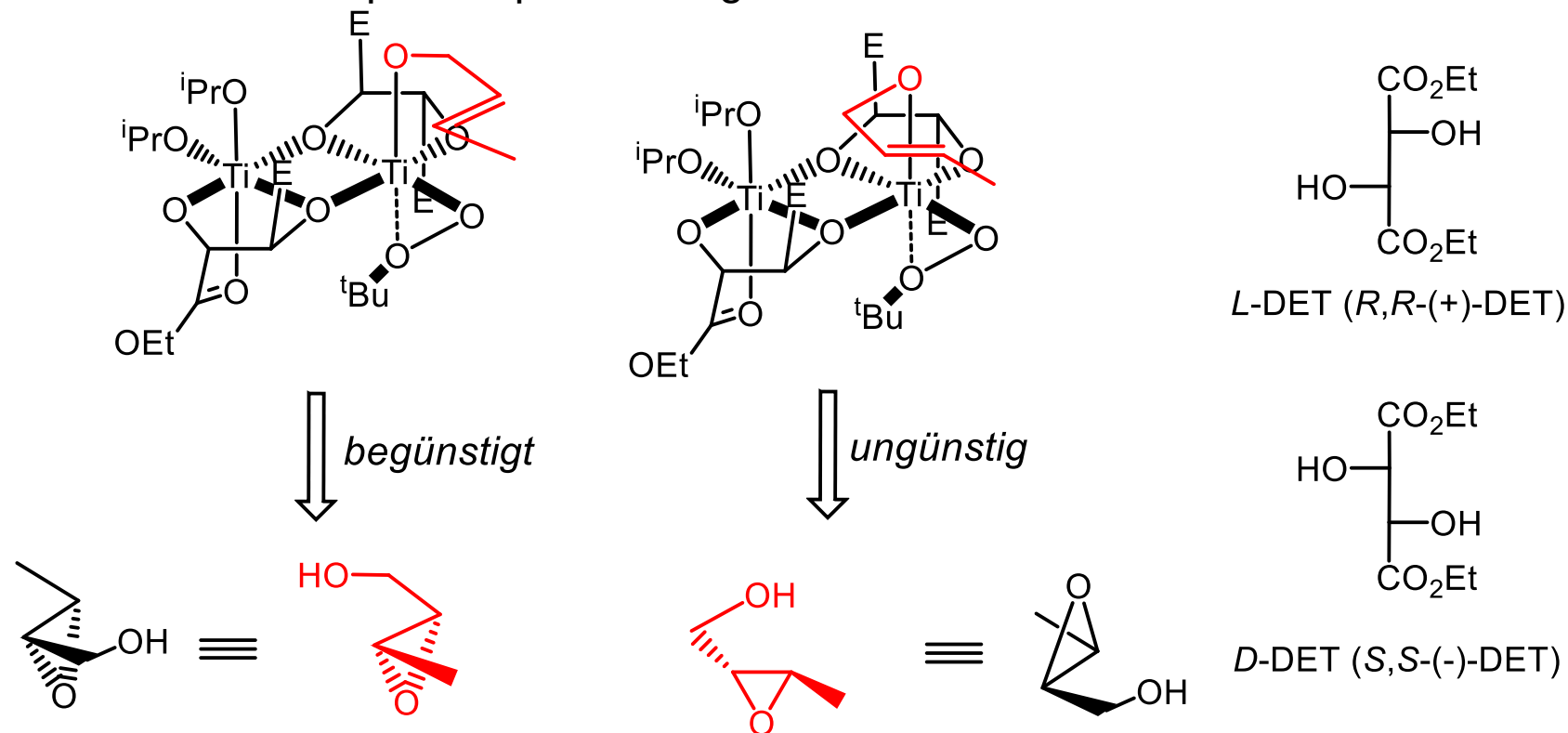
Oxo-Komplex
Jacobsen-Epoxidierung

Peroxy M = Ti, V	Peroxo M = Re	Oxo M = Mn, Cr

3.1.2.1. Oxiran

Fortsetzung

Enantioselective Sharpless-Epoxidierung

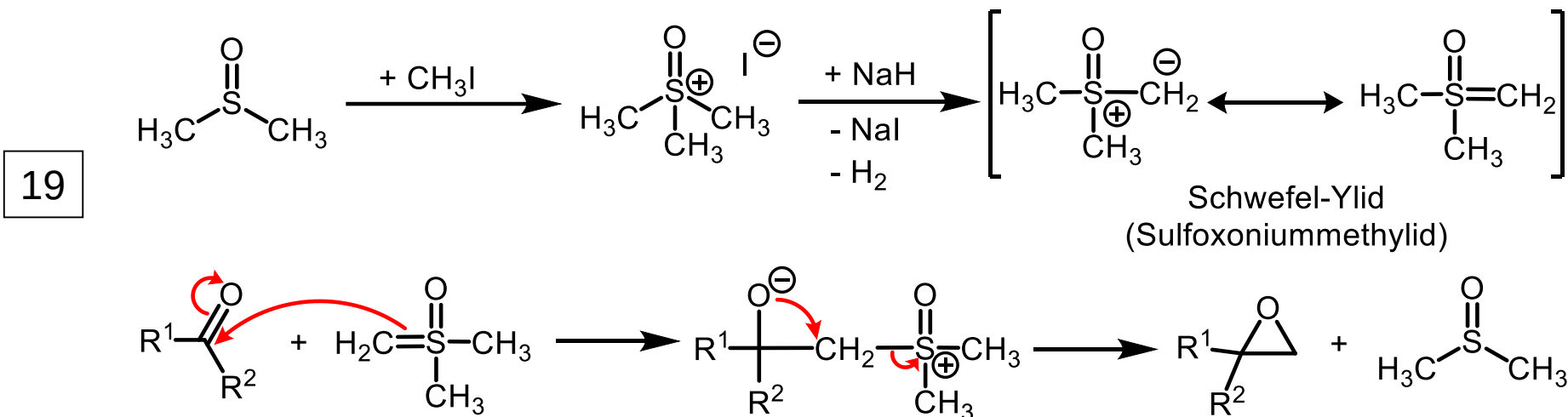


18

3.1.2.1. Oxiran

Fortsetzung

d) Corey-Synthese



e) Epoxidierung von Alkenen mit Dioxiranen

Dimethyldioxiran (DMDO):

Murray, *JACS* **1984**, 2462;

JOC **1985**, 2847.

Adam, *JOC* **1987**, 2800.

Trifluordimethyldioxiran (TFDO):

Curci, *JOC* **1988**, 3890.

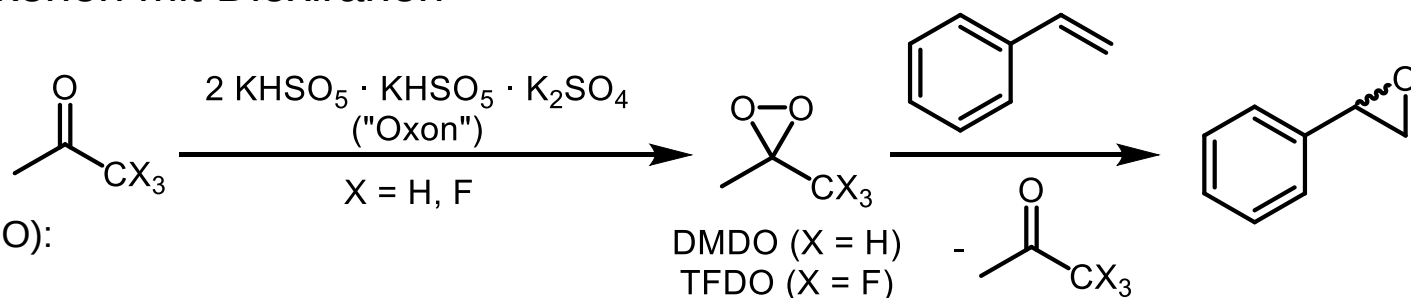
Reaktivität gegenüber Alkenen:

TFDO (100000) >

DMDO (100) >

Peroxybenzoesäure (1)

Heterocyclen



Ringspannung:

DMDO: ca. 75 kJ/mol

Dioxiran: ca. 105 kJ/mol

TFDO: ca. 110 kJ/mol)

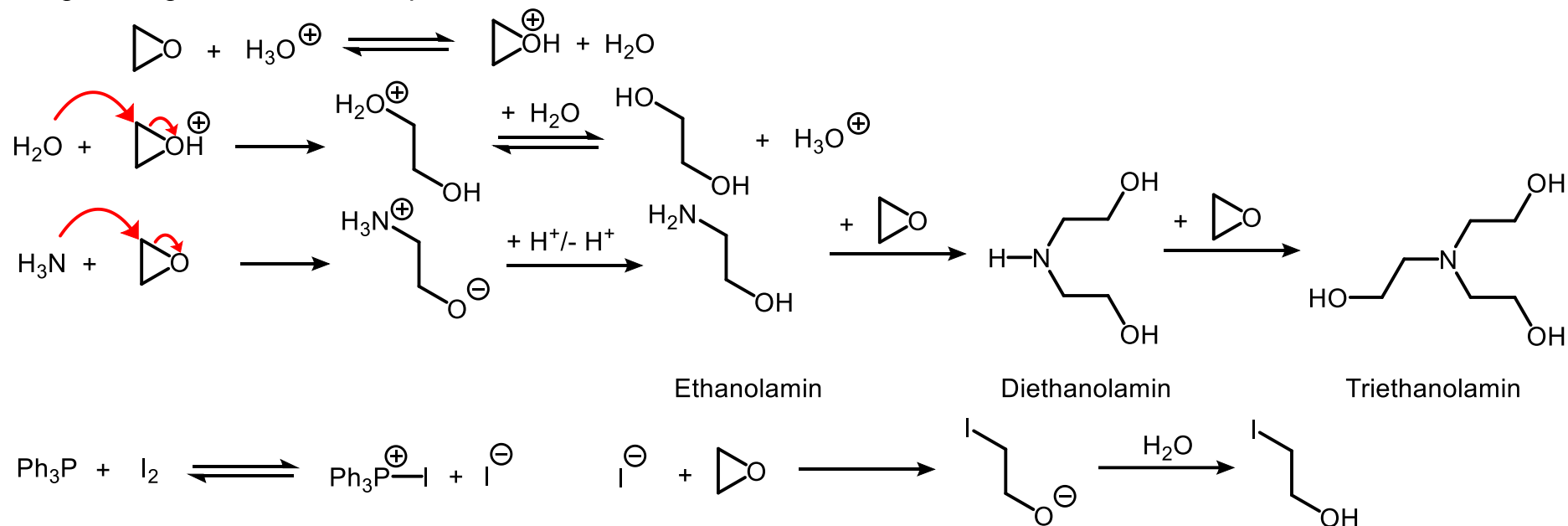
Oxiran: ca. 115 kJ/mol

- Triebkraft: Bruch der O-O-Bindung
- O-Elektrophilie entscheidend für die RG
- Ringspannung ist irrelevant

3.1.2.1. Oxiran Fortsetzung Reaktionen

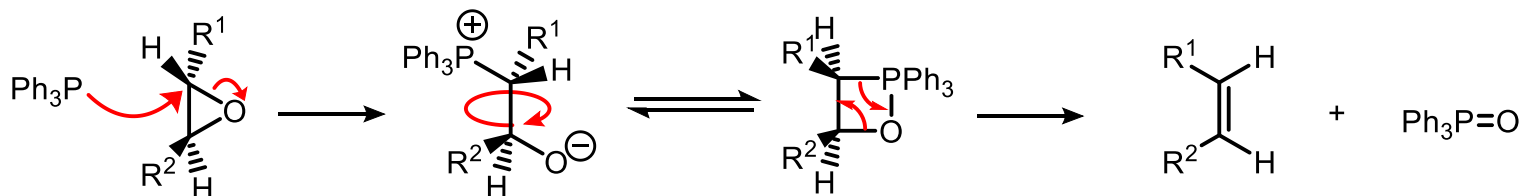
20

Ringöffnungen durch Nucleophile



21

Deoxygenierung zu Olefinen



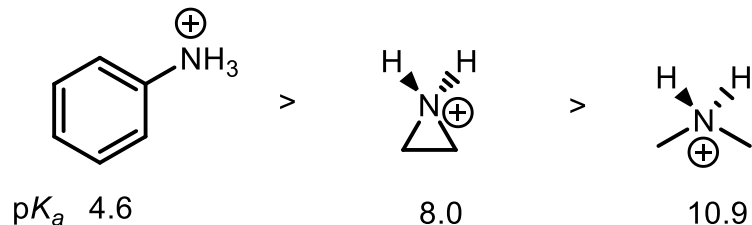
durch Epoxidierung
von *trans*-Olefin

Heterocyclen

3.1.2.2. Aziridin

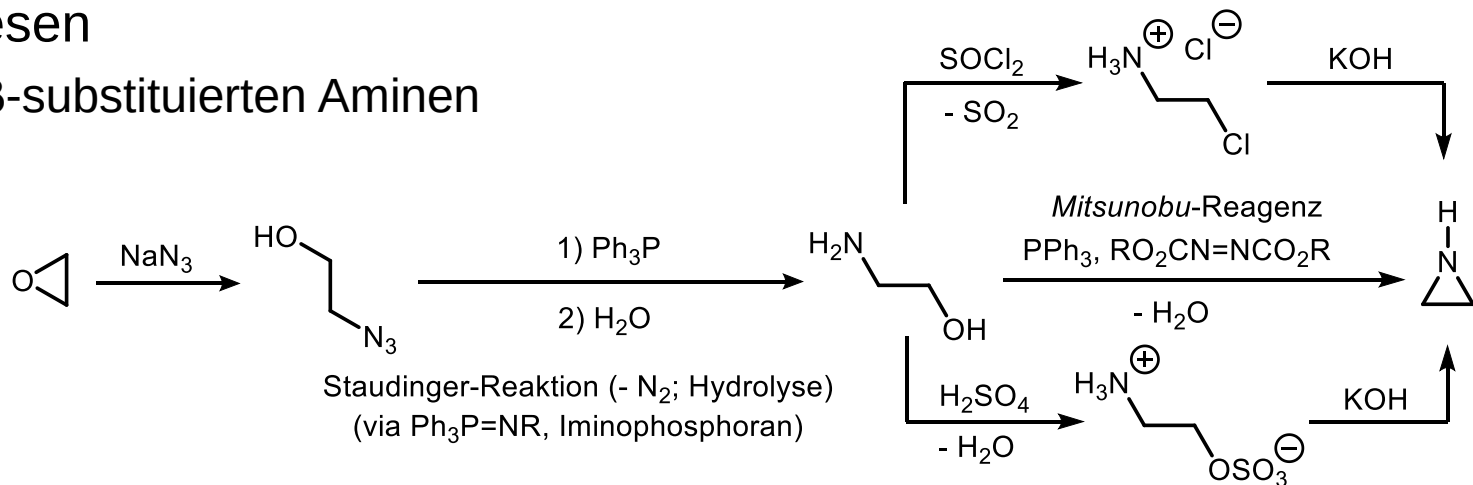
Basizität (Acidität der konjugierten Säure)

22

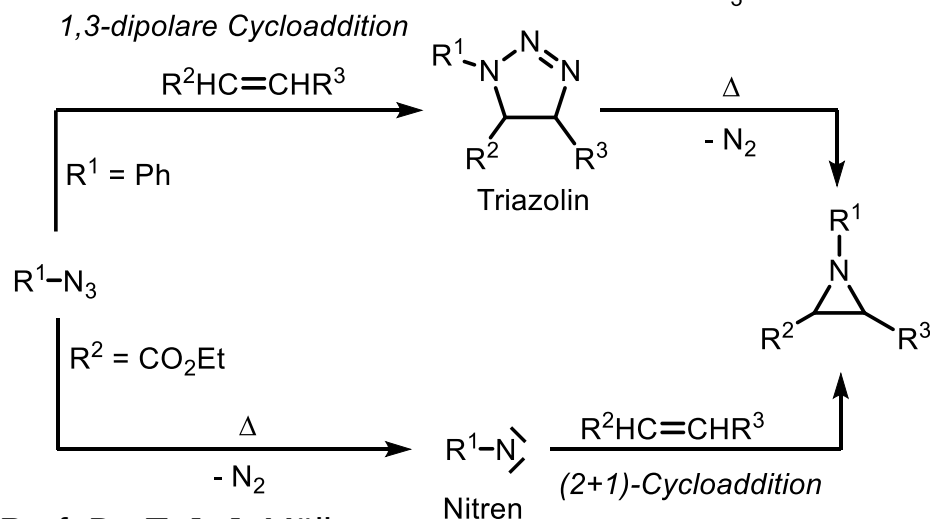


Synthesen

a) Aus β -substituierten Aminen



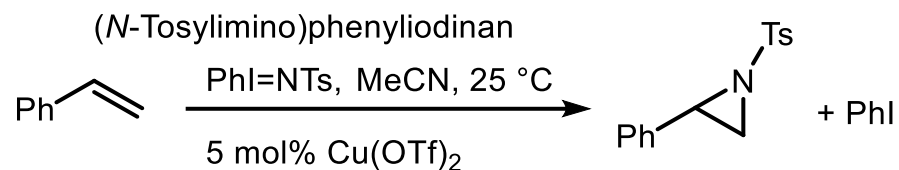
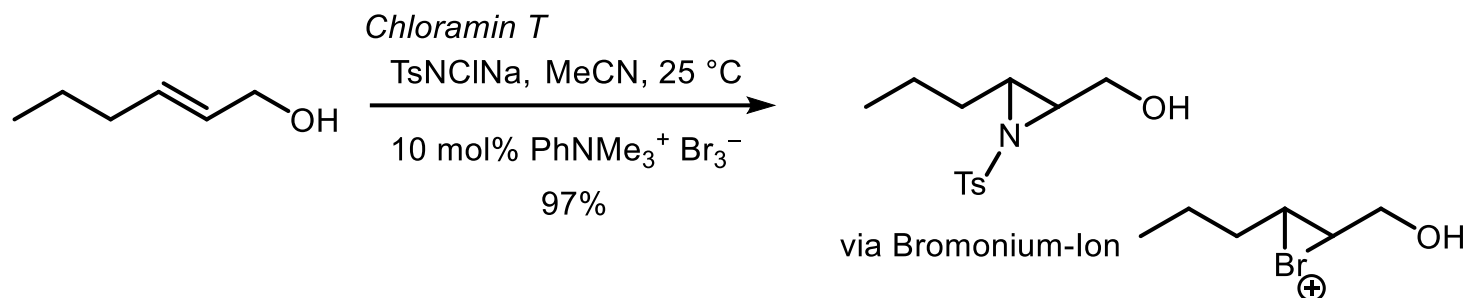
b) Aus Olefinen und Aziden



3.1.2.2. Aziridin

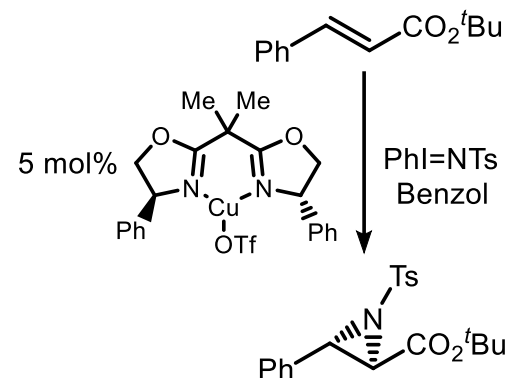
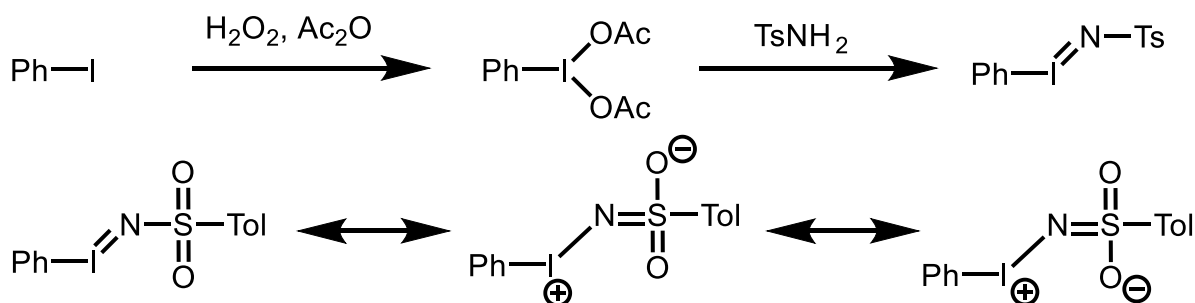
Fortsetzung

c) Aus alternativen Nitrenquellen



Evans et al., *JOC* **1991**, 6744; *JACS* **1994**, 2742; Dodd et al., *Synlett* **2003**, 1571.

Fe(III)-, Mn(III)-katalysiert: Mansuy et al. *JCSCC* **1984**, 1161.



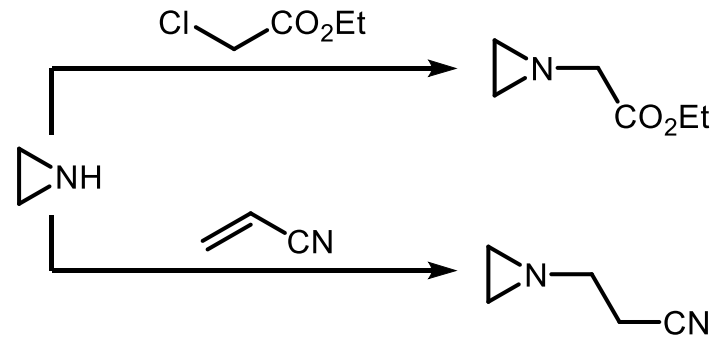
Bisoxazolin-Cu-Komplexe: Evans, *JACS* **1993**, 5328.
 Salen-Cu-Komplexe: Jacobsen, *JACS* **1993**, 5326.

3.1.2.2. Aziridin

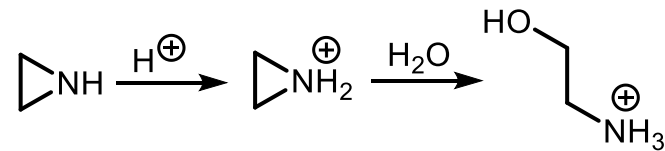
Fortsetzung

Reaktivität

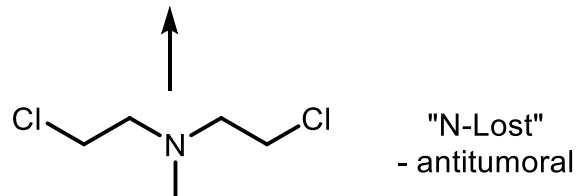
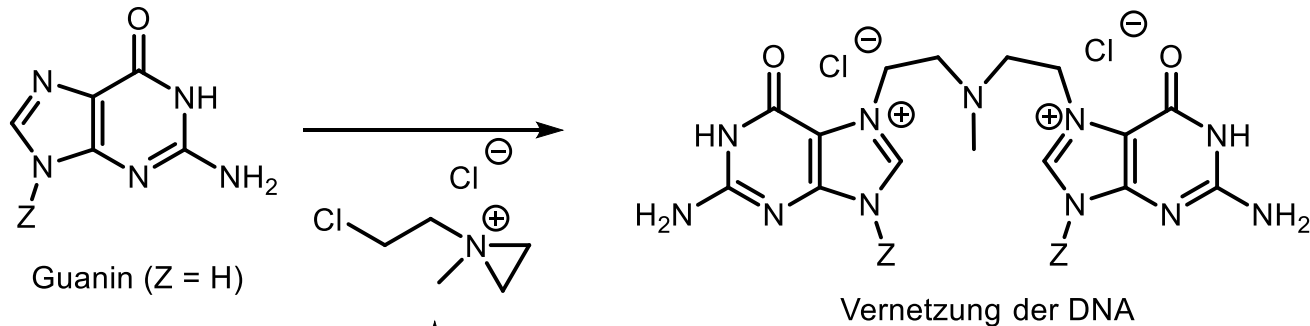
a) Gegenüber Elektrophilen



b) Elektrophilkatalysierte Ringöffnung durch Nucleophile



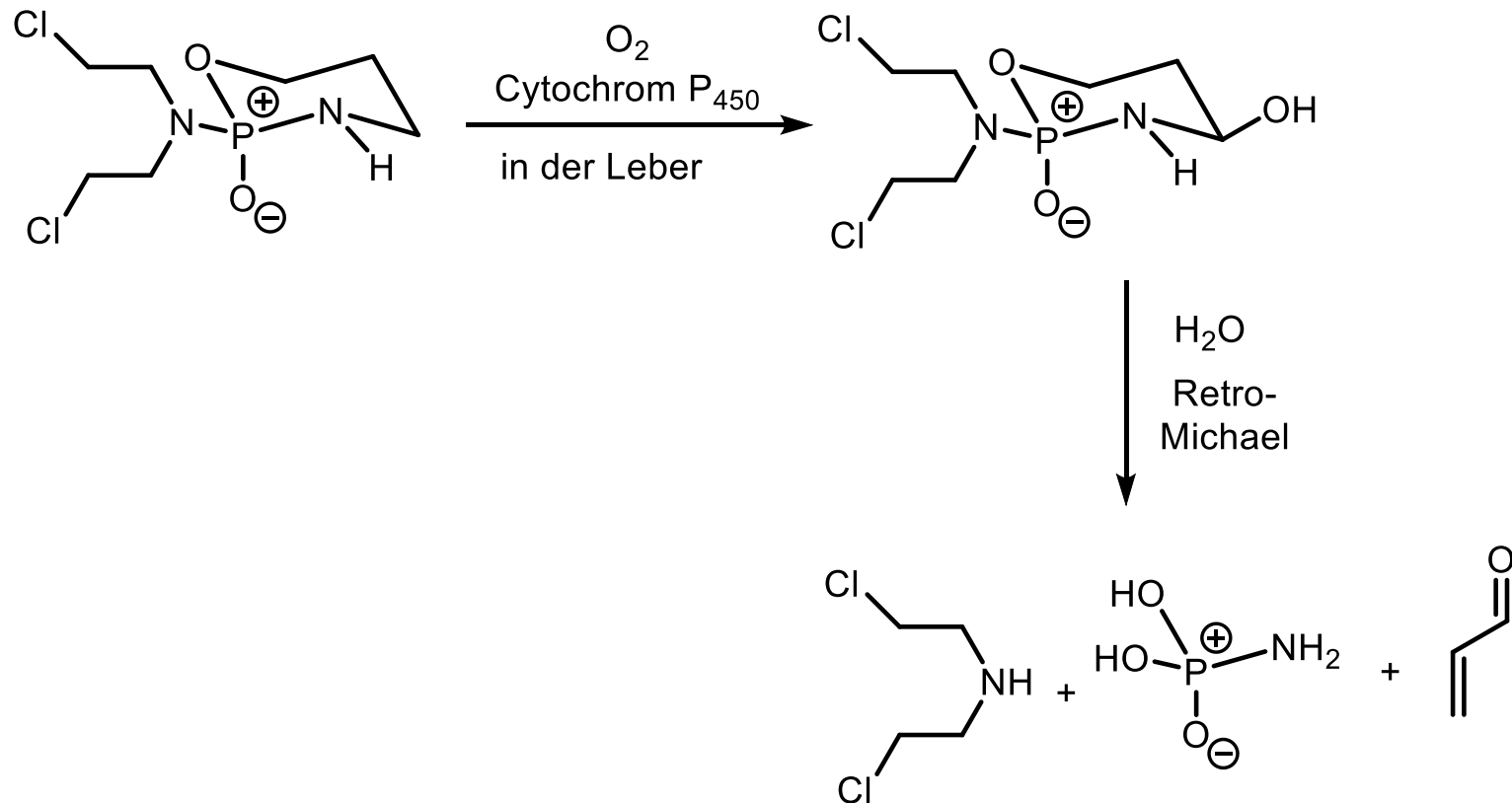
z. B. Schädigung von Nucleinsäuren



3.1.2.2. Aziridin

Fortsetzung

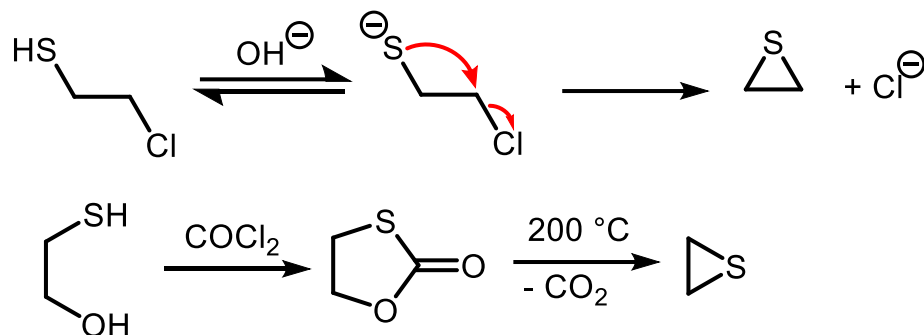
Therapeutikum Cyclophosphamid (maskierter N-Lost)



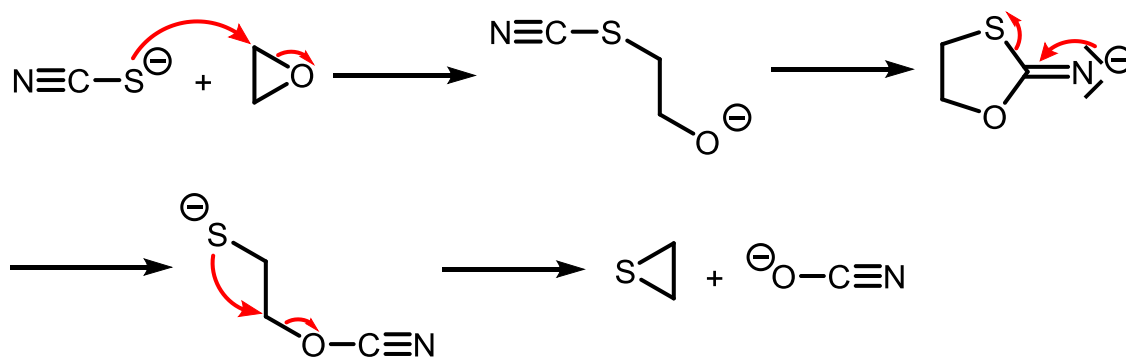
3.1.2.3. Thiiran

Synthesen

23 a) Cyclisierung von β -substituierten Thiolen

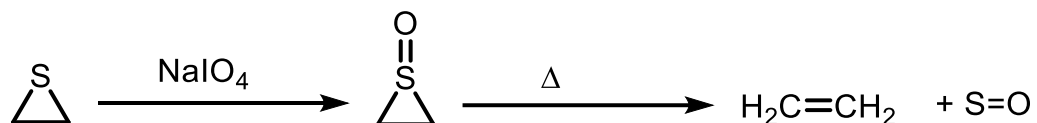


b) "Ringtransformation" von Oxiranen

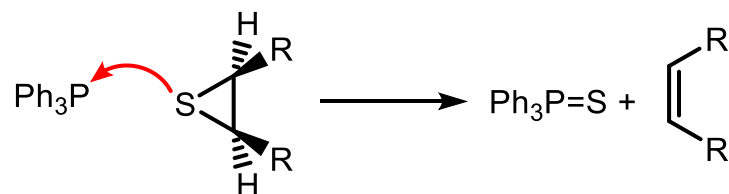


Reaktionen

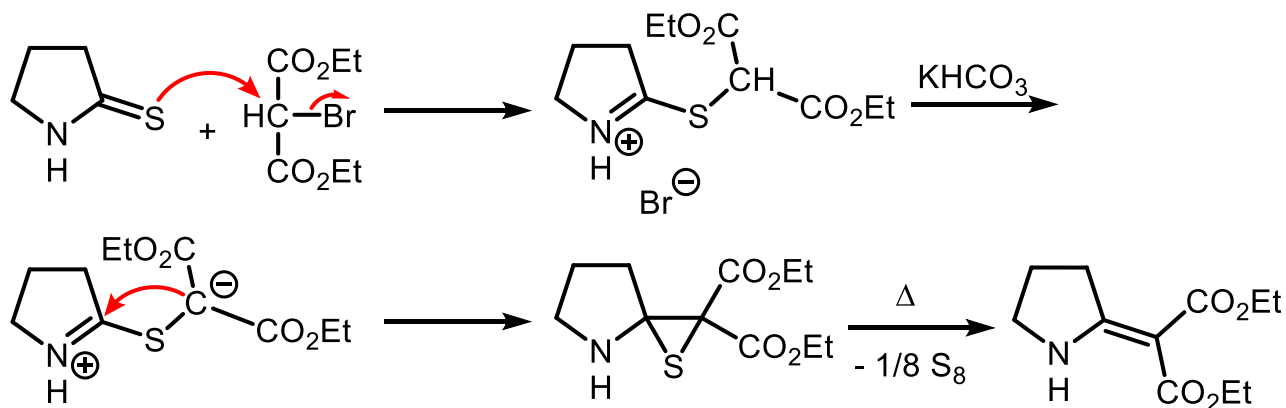
a) Oxidation zu S-Oxiden



b) Desulfurierung zu Olefinen



c) Sulfid-Kontraktion nach Eschenmoser

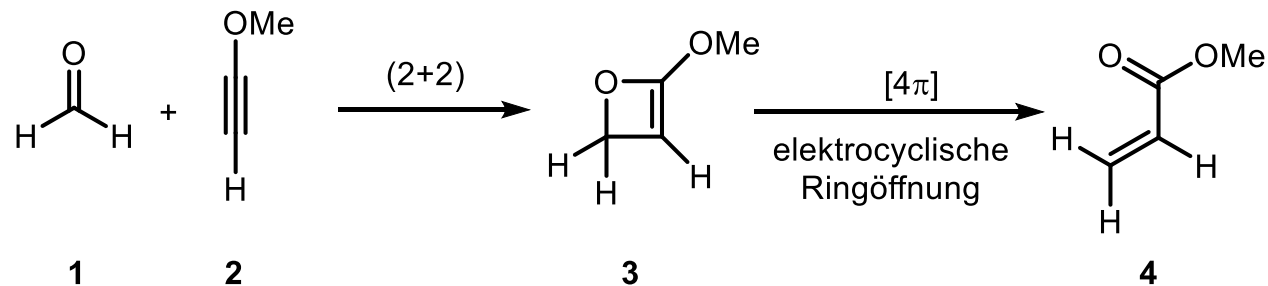
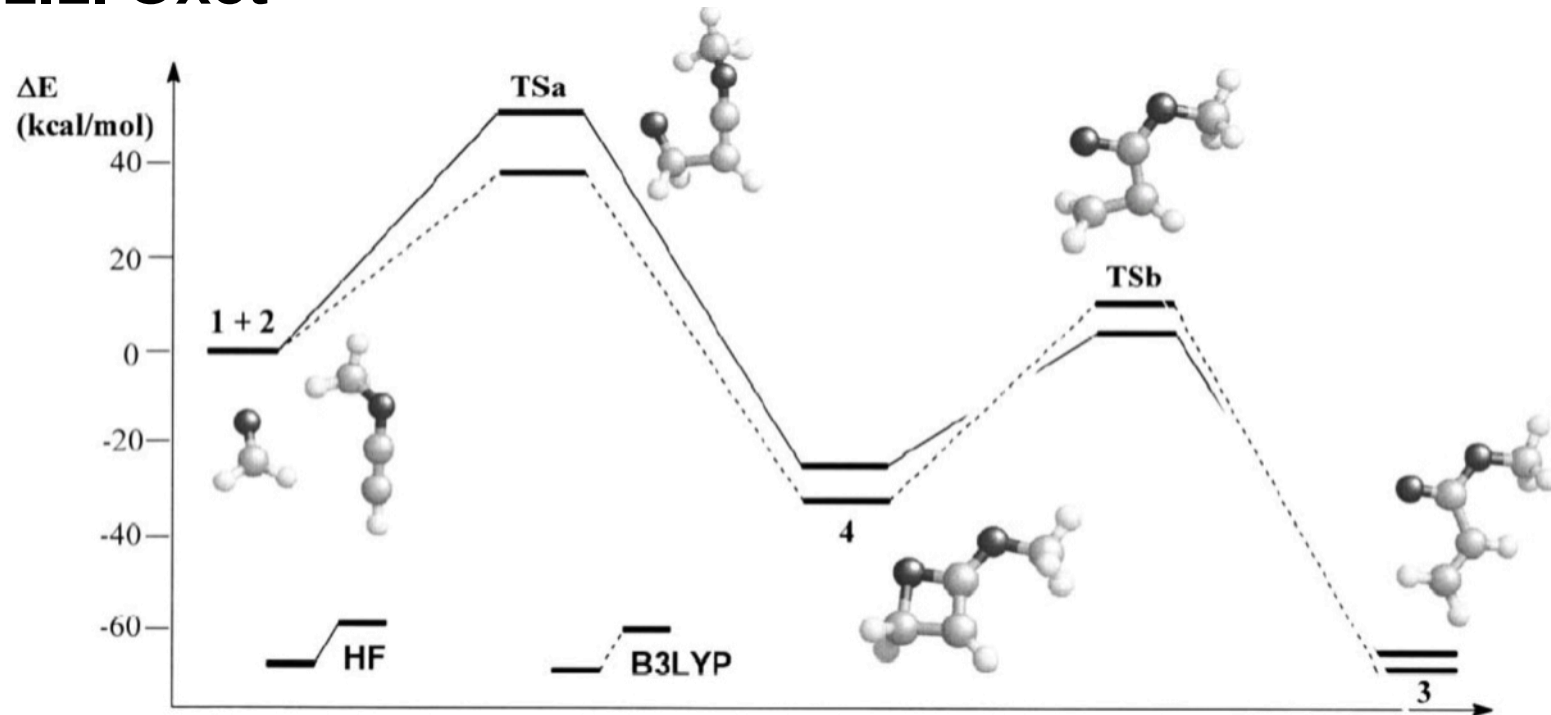


24

3.2. Vierring-Heterocyclen

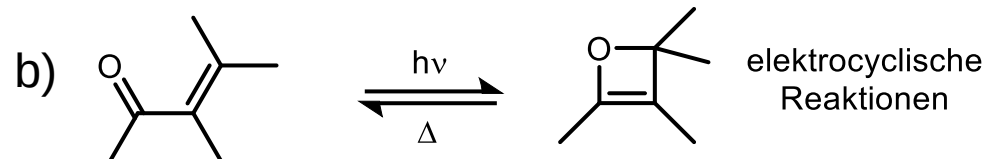
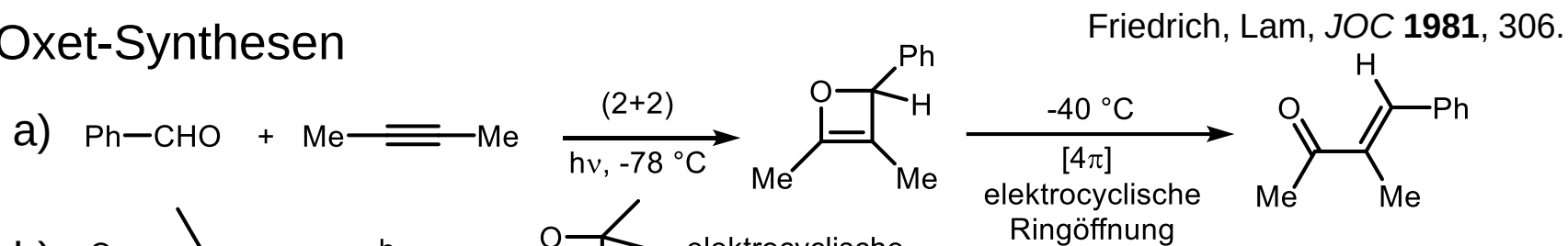
3.2.1. Ungesättigte Vierringheterocyclen

3.2.1.1. Oxet

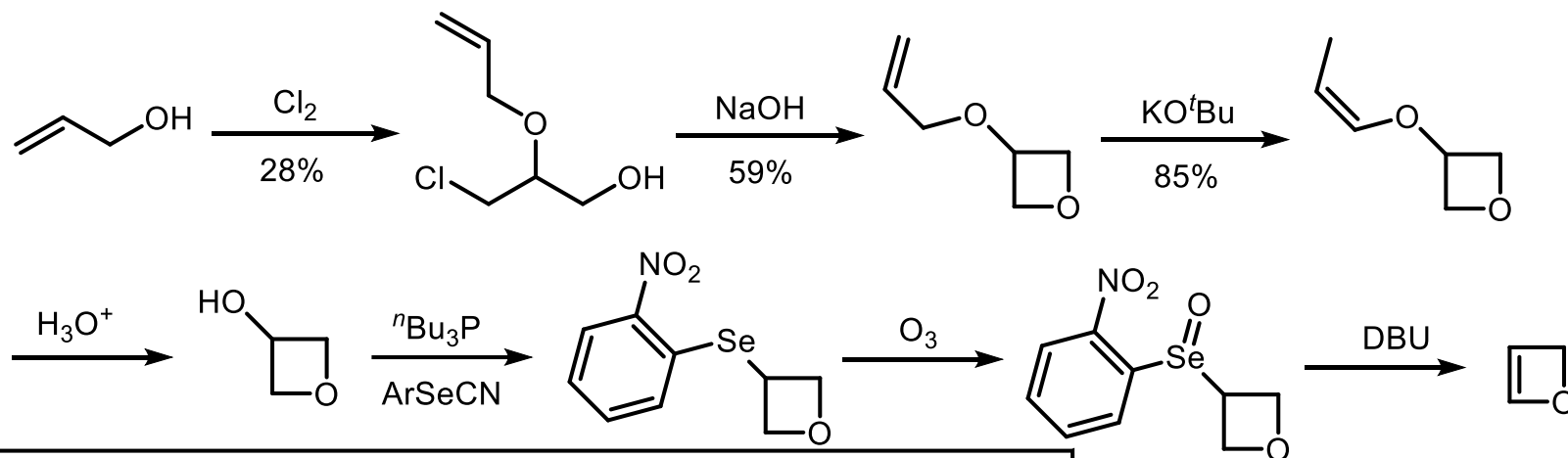


3.2.1.1. Oxet Fortsetzung

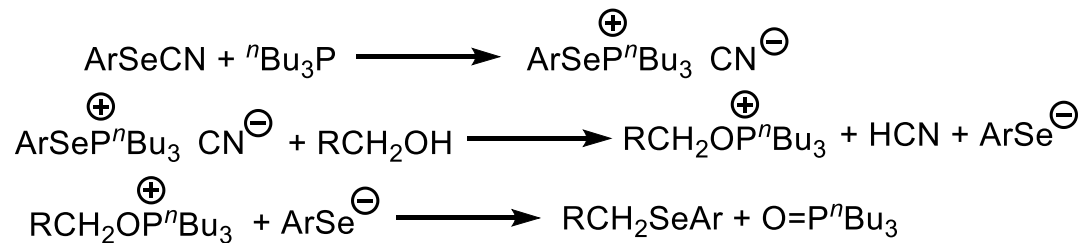
Oxet-Synthesen



c) Stammverbindung Grieco, JOC **1976**, 1485.

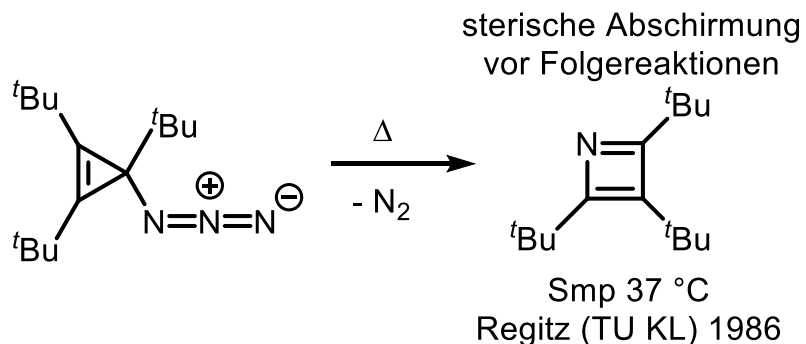


25

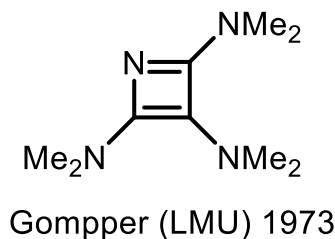


3.2.1.2. Azet

 isoelektronisch zu Cyclobutadien (4π -Antiaromat); Stammverbindung unbekannt

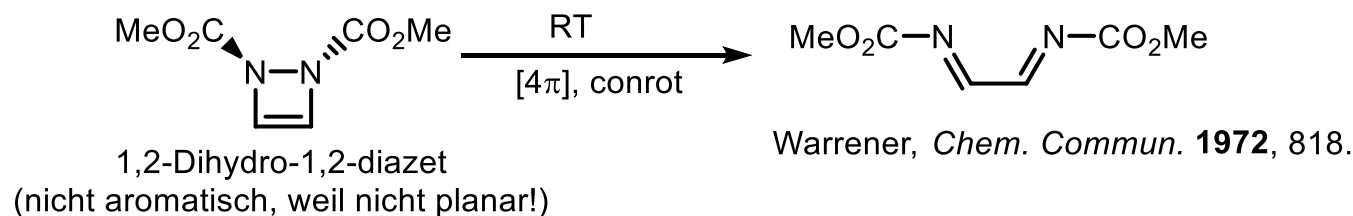


elektronische Absenkung
durch Donorsubstitution



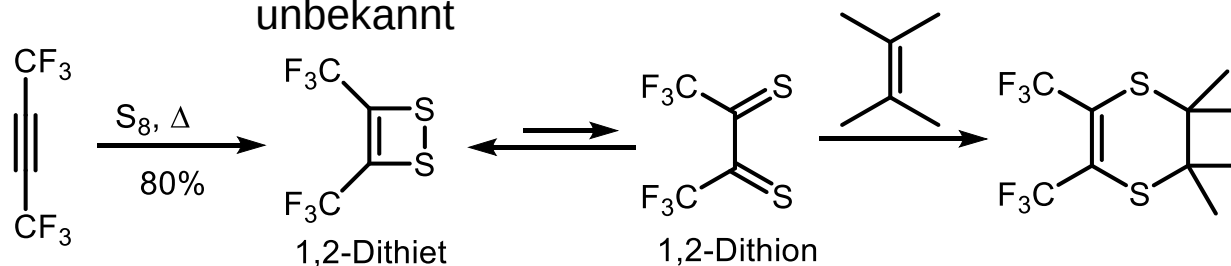
(2+2)-Cycloreversion thermisch verboten nach Woodward-Hoffmann-Regeln

3.2.1.3. 1,2-Dihydro-1,2-diazet



3.2.1.4. 1,2-Dithiet

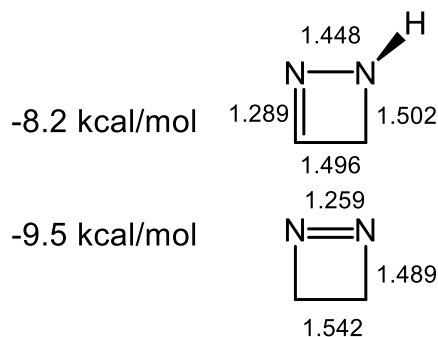
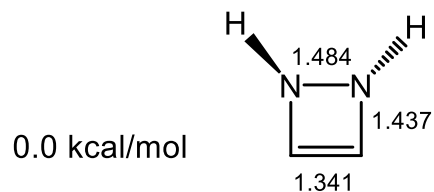
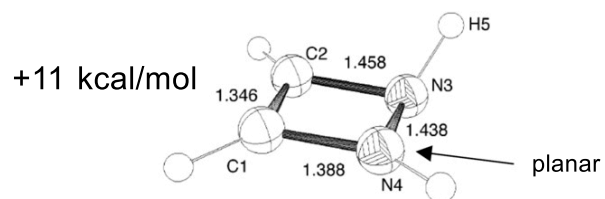
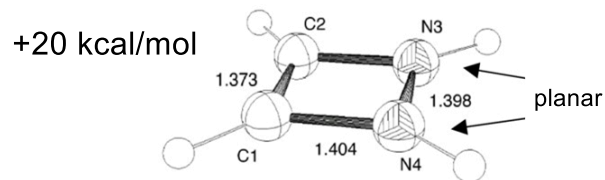
isoelektronisch zu Benzol (6π -Aromat), Stammverbindung trotzdem unbekannt



3.2.1.3. Dihydrodiazet/3.2.1.4. Dithiet Fortsetzung

Einschub: Sind 6π -Vierringe aromatisch?

Energie



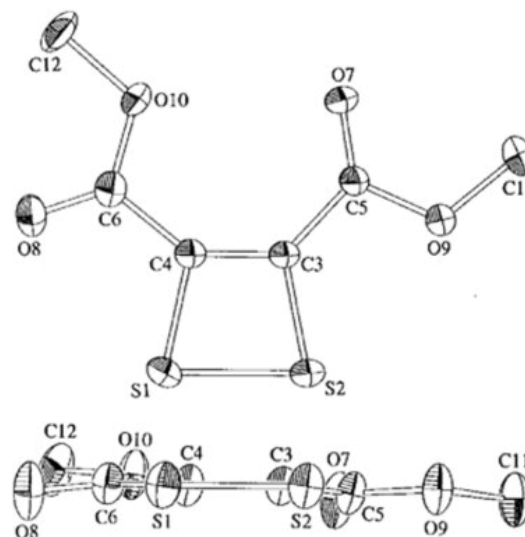
Heterocyclen

Breton, *JOC* **2002**, 6699; Budzelaar, *JACS* **1987**, 6290

Theoretische Studie auf DFT-Niveau zur Stabilität der diversen Isomere (RB3LYP/6-311+G(2d,p))

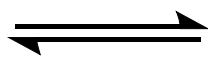
⇒ 1,2-Dihydro-1,2-diazet ist nicht aromatisch

1,2-Dithiet ist im Kristall planar.
(Kamigata, *JOC* **1998**, 6192.)

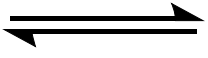


3.2.2. Gesättigte Vierringheterocyclen

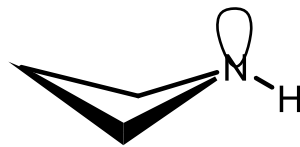
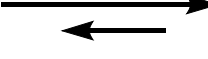
Oxetan, Thietan, Azetidin



Ringspannung: 106.3 kJ/mol
Inversionsbarriere 0.18 kJ/mol $\Rightarrow$ geringer
als die Schwingungsenergie $\Rightarrow$ rasche
Inversion



Inversionsbarriere 3.28 kJ/mol $\Rightarrow$ oberhalb
der niedrigsten 4 Schwingungsniveaus $\Rightarrow$
langsamere Inversion $\Rightarrow$ Ring ist nicht
planar



Inversionsbarriere 5.5 kJ/mol
(vgl. Cyclobutan: 6.2 kJ/mol)

Verringerung der Pitzer-Spannung durch Faltung ("puckering"). Spannungsenthalpie ist nur 3-7 kJ/mol geringer als die der Dreiring-Heterocyclen.

Gesättigte Vierringheterocyclen sind wie die Dreiring-Heterocyclen durch S_Ni (Cyclisierung) der 3-Halogenverbindungen herstellbar.